Gravity Tied to Dark Energy: The Acceleration Law, a Cold Dark Energy Sector, and a Core Law from Galaxies to Clusters

Flagship working note (PAPER61), 2026-10-25, v1.0 (draft 1, for the owner's review; not deposited). AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted; the theory is not closed.

Abstract

This programme studies gravity in which the galactic acceleration scale is set by dark energy, $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ with $\kappa = \frac{1}{2}$ fitted (equivalently $G\rho_\Lambda = 4a_0^2$, $\Lambda = 32\pi a_0^2/c^4$), and in which the gravitating mass that baryons do not supply is a collisionless *cold dark energy*: a field component with $w \approx 0$, paired with dark energy, not a particle. Following the owner's decision of 2026-10-25, this note presents the theory as the acceleration law plus a cold dark energy sector whose halo cores are set by a_0. Every result below comes from a lane that froze its criteria before computing and ran a mutation control, on both a_0 footings (9.36×10^{-11} and $1.13 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, never pooled; pairs are canonical / alternative).

1. **The law.** At galaxy scale the law passes:
   - the SPARC radial-acceleration relation (rms 0.054 dex);
   - MeerKAT's local a_0;
   - isolated-disc weak lensing ($A = 1.016 \pm 0.086$);
   - a screened Solar System.

   Its a_0 tracks dark energy: the native-ruler BAO reconstruction gives $a_0(z)/a_0(0)$ in [0.76, 1.03] at $z = 1$ (68%).
2. **The cold sector, closed doors.** Twelve mechanisms that would add, heat, select or redistribute cold dark energy were tested and closed. These include hot-gas tracking (the measured sign is opposite), any dark-energy-keyed heater (a certified no-fold theorem), feedback, baryonic contraction, and member-halo nesting.
3. **The cold sector, the core law.**
   - *The anchor.* Halos cored at the surface density $\Sigma_{a_0} = a_0/(2\pi G)$ (107 / 129 $\mathrm{M_\odot\,pc^{-2}}$) pass SPARC. Their pull is certified never to exceed $\frac{9}{32}a_0$ at any radius.
   - *The law.* With a self-coupled cold dark energy the core law is $\Sigma_0^2 = 9\sigma^{1+n}/(4\pi GKt)$ (Lean-certified). Anchoring K by Σ_{a_0} at the median galaxy and fixing the one exponent on SPARC alone ($q_f = 0.892$ / 0.748) improves SPARC to rms 0.0605 / 0.0591.
   - *Out of sample.* It predicts 14 of 15 cluster cells of 326 eRASS1 clusters that were never fitted. The heavy-core misfit falls from $\sum |Z| = 17.61$ to 5.31 (canonical).
   - *Physics reading.* The implied self-interaction falls with velocity, $n = 0.784$ / 0.496.
4. **Open.** The same law fails the faint dwarfs (48–65% exceed their cap). Rate saturation, gravothermal collapse and a self-consistent halo dispersion are each ruled out as the fix. The dwarf excess behaves like a separate small-scale floor. The innermost cluster band is mildly over-predicted. No native CMB or growth computation exists yet for the self-coupled sector.
5. **Certified mathematics.** Sixteen theorems (Lean 4 + Mathlib, no `sorry`, standard axioms only, mutation copies fail to compile) carry the core-law chain from the Burkert mass to the dwarf tests.

1 Framework, rules and the owner's decision

- **The law.** $a_0 = \kappa c\sqrt{G\rho_\Lambda}$; $\kappa = \frac{1}{2}$ is FITTED, not derived. Of 537 derivation attempts on the record, none derives it (the κ ledger). Footings: canonical $a_0 = 9.3603 \times 10^{-11}$, alternative $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, never pooled. Status words follow the record: DERIVED, COMPUTED, POSTULATED, FITTED, OPEN, PENDING.
- **The decision.** On 2026-10-25 the owner chose to present the theory as the acceleration law plus a cold dark energy sector whose cores are set by a_0 (route E: the cold sector gravitates normally and the galaxy law is emergent from it). The covariant Khronon theory v3s (PAPER58–60) is kept as the covariant attempt: it passes as a galaxy law and fails as a theory of clusters, growth and the early universe (PAPER60).
- **The discipline.** Each lane committed its frozen criteria alone before any script or number, ran a mutation (MUTATE) control, and reported amendments. Lanes live in `rebuild_2026_10_22/`. Nothing here is evidence for the framework over ΛCDM or any rival.

Table 1: Closed doors for the cold sector. Each lane froze its criteria first; commits are result commits.

mechanism	lane	verdict	commit
one selector variable	X19	no single variable separates	a9ec701f5b
universal late heating	X23	WRONG AMPLITUDE	d5f2a449d1
vacuum heating $Q = \epsilon H \rho_\Lambda c^2$	X24b	NO EPS WINDOW	3492bcbeb4
isothermal partial-retention sheet	X26, X30	PROFILE EXCLUDES THE SHEET	0f19b6759b
feedback energy	X31	FEEDBACK FAILS	5695a61801
orbit-keyed conversion	X37	FAILS CLUSTERS	36b7be3e67
baryonic contraction	X39	FAILS	54684fb62f
single-halo Σ rules	X40b	NO RULE CLOSES IT	ba368d2ba1
hot gas as selector	HG1, HG2	UNDERPOWERED; $D = -0.294$, opposite sign	b58968185a
dark-energy-keyed heater	DH1	OBSTRUCTION	3870d2c4e8
member-halo nesting	NM1	NM FAILS	408e26a409
seven named dwarf floors	DF1	NONE SURVIVE	4513c7c324

2 The acceleration law

The law is the simple MOND relation on baryons, $g_N = g^2/(g + a_0)$, with a_0 tied to the dark-energy density. In its own regime of isolated systems it passes:

- **Rotation curves:** the SPARC radial-acceleration relation, rms 0.054 dex on both footings (P4; EM1 reproduction 0.0541 / 0.0537).
- **The local scale:** MeerKAT's a_0 sits +0.032 dex from SPARC's with the same recipe (PAPER60).
- **Isolated-disc lensing:** KiDS-1000 blue lenses on the law, $A = 1.016 \pm 0.086$ (CFG829; reproduced in HG2 as 1.0165 ± 0.0859).
- **The Solar System:** screened, Cassini passes (PAPER58).
- a_0 **tracks dark energy.** With the native sound horizon, DESI BAO plus Pantheon+ reconstruct $\sqrt{X} = a_0(z)/a_0(0)$ in [0.76, 1.03], [0.71, 1.12] and [0.32, 1.21] at $z = 1$, 1.5 and 2.33 (68%). The rival $a_0 \propto H(z)$ is far outside these intervals. A constant dark energy is not rejected (1.97σ, class CONSISTENT WITH CONSTANT; CFG689, 6c064b83a3). The decisive point is $z \approx 2.5$ at 0.10 dex.

The law does not, by itself, explain the cluster cores, the faint dwarfs or the growth of structure. Those need the cold sector.

3 What the cold sector must do

The committed specification (`COLD_SECTOR_SPEC_2026-10-24.md`) lists six requirements, R1–R6:

- **R1** dust-like from recombination at $\Omega_c/\Omega_b = 5.364$;
- **R2** clustering like dust down to forest scales through $z \approx 2$–3;
- **R3** almost absent from isolated galaxies' inner regions today;
- **R4** present in cluster cores;
- **R5** a graded extra in faint dwarfs;
- **R6** collisionless at merger speeds (the Bullet Cluster).

The cluster requirement is model-free: at matched baryonic acceleration, seven X-COP clusters sit +0.416 dex above the median SPARC disc (the matched-clumping lane, c51d962827). Their excess is more concentrated than the hot gas.

Two of these closures are theorems (Section 6).

- **A uniform drive cannot heat.** A spatially uniform drive (anything keyed to dark energy) drops out of the fold source $\partial^2 H/\partial x^2 = \Phi''$, so it cannot heat a cold stream.
- **The heating windows do not overlap.** If heating per unit mass scales with σ^2, then $c_s = A\sigma$ with one A. SPARC needs $A \geq 2.695$ and the clusters allow $A \leq 1.424$; no A does both.

HG2 found that at matched σ, X-ray-bright groups show *less* excess than X-ray-faint ones once their gas is counted: $D = -0.294$ dex, 95% interval $[-0.511, -0.076]$.

Table 2: OS1, the core law out of sample. Clusters: X38's 15 cells on 326 eRASS1 clusters, both h. Z shown at $h = 0.672$.

	canonical	alternative
q_f (SPARC only)	0.892	0.748
SPARC rms @ median Υ (X38)	0.0605 @ 0.79 (0.0795)	0.0591 @ 0.63 (0.0729)
cluster cells passing	14 of 15	14 of 15
failing $\sum \lvert Z \rvert$, X38 $\rightarrow$ OS1	17.61 $\rightarrow$ 5.31	14.26 $\rightarrow$ 4.40
MID T3 Z (X38 +4.26)	-1.29	-0.89
OUTER M3 Z (X38 +5.36)	$+1.81$	$+1.99$
INNER M-all Z (the one miss)	-2.41 (over)	-2.06 (over)
dwarfs at $z \geq 2$, median S	65.2%, 4.78 (EXCLUDE)	48.3%, 2.74 (EXCLUDE)

4 The core law

4.1 Cores at the a_0 surface density

In route E the cold dark energy gravitates normally and forms cored (Burkert) halos. X38 (f72216ad98) set the core surface density to

$$\rho_0 r_0 = \Sigma_{a_0} = \frac{a_0}{2\pi G} = 107 \ / \ 129 \ \mathrm{M_\odot \, pc^{-2}},$$

with no new constant.

- **SPARC:** passes, rms 0.0795 / 0.0729 at median Υ 0.77 / 0.60.
- **Cluster OUTER bands:** pass. Every earlier added-component model overshot them.
- **Fails:** the cores of the heaviest, hottest clusters.

The anchor is comparable to the empirical constant core surface densities of disc galaxies (Donato et al. 2009); that is context, not a test. A fixed-Σ Burkert halo's pull is $GM/r^2 = \pi G \Sigma \, \mathrm{Bk}(x)/x^2$ with $x = r/r_0$, so over every core radius it is capped: certified $\leq \frac{9}{32} a_0$, numerically $0.254484 \, a_0$ (LN1). On X1's 89 dwarfs per footing this cap is CONSISTENT: 1.1% exceed it at $z \geq 2$, with median required fraction $S = 0.53$ / 0.44 (CD1, 809e61b319).

4.2 A self-coupled cold dark energy

Let the cold dark energy interact with itself, with cross-section per mass $\propto v^{-n}$, and let the core form where rate $\times$ age $= 1$ on an isothermal core. Then

$$\Sigma_0^2 = \frac{9 \, \sigma^{1+n}}{4\pi G K t}, \qquad \Sigma_0 \propto \sigma^q t^{-1/2}, \quad q = \frac{1+n}{2}$$

(LN1, theorem T7). Fixing K so that the median SPARC galaxy ($\sigma_{\mathrm{ref}} = 70.50 \, \mathrm{km \, s^{-1}}$, $V_{\mathrm{last}}/\sqrt{2}$) keeps Σ_{a_0} makes a_0 set the coupling's normalisation:

$$\Sigma_i = \Sigma_{a_0} \, (\sigma_i/\sigma_{\mathrm{ref}})^q.$$

At the fixed candidate $q = \frac{1}{3}$, SPARC improves on X38 (rms 0.0686 / 0.0636) and the clusters' failing $\sum \lvert Z \rvert$ falls from 17.61 to 11.40 with no new failure (SC1, d2f4a621d8; frozen class SC PASSES SPARC ONLY).

4.3 Out of sample: one constant from galaxies predicts the clusters

OS1 (frozen 71873e76d3; result a8b817e19e) calibrated q once on SPARC alone and graded clusters and dwarfs at that q with no further freedom (Table 2).

The frozen class is OS FAILS OUT OF SAMPLE, because the dwarfs fail and one cluster cell over-predicts. The cluster result itself is the strongest on the record: a single constant fixed on galaxies turns X38's five failing cluster cells into passes. The MUTATE control permutes σ across the SPARC galaxies before calibrating, and the SPARC gain collapses from 0.0190 to 0.0003 dex. The gain therefore comes from the real σ ordering.

4.4 Physics reading

- **The self-interaction falls with velocity.** $q = (1+n)/2$ gives $n = 0.784$ / 0.496 (LN2). The cross-section per mass falls with velocity, the conventional direction for velocity-dependent self-interacting dark sectors.
- **What differs from such a sector.** It differs in two places: the normalisation is fixed by a_0, hence by Λ, and the component is a field paired with dark energy, not a particle species.
- **It never reaches the fluid regime.** With the coupling anchored this way, the Knudsen number is certified to be

$$\mathrm{Kn} = \frac{4}{3\sqrt{\pi}} \frac{2a_0 t}{\sigma_{\mathrm{ref}}} \Big(\frac{\sigma}{\sigma_{\mathrm{ref}}}\Big)^{q-1},$$

  which evaluates to 653–1158 (canonical) and 539–2048 (alternative) from 5 to 1000 $\mathrm{km\,s^{-1}}$. The sector is in the long-mean-free-path regime everywhere.

5 The dwarfs: the open problem

Below $\sigma \approx 20\,\mathrm{km\,s^{-1}}$ the core law gives cores too shallow for the faint dwarfs, which need 2–10 times more central pull. DW1 (279b4aa47c / 0dd2bffffa) tested three stated mechanisms:

- **M1, rate saturation:** RULED OUT ($\mathrm{Kn} \gg 1$).
- **M2, gravothermal collapse:** RULED OUT. Under the core law $\Gamma t = 1$ in every system, a certified structural statement.
- **M3, the halo's own dispersion:** FAILS. The certified unique solve gives σ_h only about twice the stellar σ; 64.0% / 40.4% of dwarfs still exceed the cap.

Independently:

- **The floor:** the dwarf extra behaves as a velocity floor of about 4–4.6 $\mathrm{km\,s^{-1}}$ (X24, 19d31a99d9).
- **Its radial form:** $g \propto r^{-0.8}$ (X28, dad0c9f1fe).
- **Its equation of state:** $\Gamma = 0.88$ $[0.81, 0.94]$, softer than isothermal, which no named equation of state supplies (DF1).

The dwarf excess needs physics that acts only at small scales; it is the second blank in the theory.

6 Certified mathematics

Two Lean 4 + Mathlib files compile with no `sorry`, and every theorem depends only on `propext`, `Classical.choice` and `Quot.sound`. Mutation copies fail to compile. Lean certifies mathematics from stated premises only. It does not certify isothermal cores, the core condition, Burkert halos, the a_0 anchor or any measurement.

- **LN1** (`ColdSectorCerts.lean`, 667db25771), nine theorems:
  - the Burkert mass derivative;
  - the fixed-Σ pull's dependence on r/r_0 alone;
  - $\mathrm{Bk}(x) \le \frac{9}{16}x^2$ and $\mathrm{Bk}(1) > \frac{1}{2}$, hence the ceiling $\le \frac{9}{32}a_0$;
  - the no-fold theorem;
  - the disjoint heating windows;
  - the distance-free baryonic acceleration;
  - the core law $\Sigma_0^2 = 9\sigma^{1+n}/(4\pi GKt)$ and its q window.
- **LN2** (`CoreLawCerts.lean`, dc2724d299), seven theorems:
  - the $q \leftrightarrow n$ map and OS1's $n > 0$;
  - the anchor-free surface-density ratio;
  - $\Gamma t = 1$ and no collapse;
  - the Knudsen closed form;
  - the unique halo-dispersion solve for $q < 2$.

7 Predictions and dated tests

- **Halo cores are set by dark energy.** A typical disc galaxy ($\sigma \approx 70\,\mathrm{km\,s^{-1}}$) has core surface density $a_0/(2\pi G)$. More generally $\Sigma_0 \propto \sigma^{0.75\text{–}0.89}$, with no further freedom. Any new sample of cored rotation curves, groups or

clusters with measured σ tests this.
- a_0 **tracks dark energy** out to $z \approx 2.5$ (target 0.10 dex; CFG689).
- **Sealed predictions on record:** ALMA (2026-11-06, 2026-11-28) and Gaia DR4 (2026-12-02), as filed under PAPER60. Those seals are for the law and v3 readings; the core law adds no sealed file yet.

8 What is fitted, open, and not claimed

- **Fitted:** $\kappa = \frac{1}{2}$; the core-law exponent q (once, on SPARC); the stellar Υ; the cold abundance $\Omega_c/\Omega_b = 5.364$.
- **Open:**
  - the dwarf floor;
  - the innermost cluster band;
  - a covariant action for the self-coupled cold dark energy and its origin in the dark-energy sector;
  - a native CMB and growth computation with the self-coupling (dust-like at early times by R1, not yet computed natively);
  - the Bullet test of the self-coupled sector (X32's reading for the hot sheet is UNDECIDED).
- **Not claimed:** a derivation of κ, a closed theory, or evidence over ΛCDM. In the core law the cold dark energy behaves at large scales like a self-interacting dust. Its distinguishing content is the a_0 normalisation and the dark-energy pairing.

Disclosures.

- **Check-definition amendments:** CD1's A1 (raw versus binary-corrected σ) and HG1's A1 (distance rescaling of the match radius) were fixed before any graded number.
- **Reporting fix:** DW1's A1 changed reporting code only, with the graded numbers identical.
- **Frozen classes as written:** SC1 is SC PASSES SPARC ONLY, OS1 is OS FAILS OUT OF SAMPLE, and HG2 is UNDERPOWERED (power 0.45 against a 0.50 bar).
- **Not the lane's own verdict:** the X19 row of Table 1 summarises that lane's finding that no single variable separates the classes.

Repository lanes: `rebuild_2026_10_22/` (X19–X40b, CD1, SC1, OS1, DW1, NM1, HG1, HG2, DH1, DF1, LN1, LN2, `COLD_SECTOR_SPEC_2026-10-24.md`); `campaign_fresh_gravity/CFG689_native_ruler_bao/`; `reviews/matched_clumping_2026_10_24/`. Companion notes: PAPER60 (doi:10.5281/zenodo.23601528), PAPER59 (doi:10.5281/zenodo.23585319), PAPER58 v1.1 (doi:10.5281/zenodo.23579379). Data: F. Lelli, S. S. McGaugh, J. M. Schombert, AJ 152, 157 (2016) (SPARC); E. Bulbul et al., A&A 685, A106 (2024) (eRASS1 clusters); D. Eckert et al., A&A 621, A40 (2019) (X-COP); M. M. Brouwer et al., A&A 650, A113 (2021) (KiDS-1000 lensing RAR); DESI Collaboration, PRD 112, 083515 (2025) (DR2 BAO); D. Brout et al., ApJ 938, 110 (2022) (Pantheon+). Context: M. Milgrom, ApJ 270, 365 (1983); A. Burkert, ApJ 447, L25 (1995); F. Donato et al., MNRAS 397, 1169 (2009). AI assistance: Anthropic's Claude and OpenAI's ChatGPT (lanes, review, drafting).